

The red eye

From: Am Fam Physician 2010;81:137, Am Fam Physician 2016;93:991, BMJ 2010;340:c1711, NEJM 2023;389:2363



Red Whale

GEMS
Guidelines & Evidence Made Simple

History and examination

History:

- Unilateral or bilateral symptoms?
- Visual changes, pain or photophobia?
- Duration of symptoms?
- Discharge?
- Headache?
- Systemic disease or allergies?
- Trauma? (penetrating/high-velocity injury or chemical)
 - Do NOT palpate the globe if perforation suspected.
- Contact lens use?
 - **ANY red eye is significant in contact lens user**
 - Risk of corneal ulceration, microbial keratitis, neovascularization
 - Infection risk increased further by: washing lenses in tap water/swimming/showering/using hot tub while wearing lenses

Examination:

- Snellen acuity, each eye separately
- Pupil size and reaction to light
- Cornea and lacrimal sac
- Fluorescein staining using a BLUE light
- Eyelids (including lid eversion)
- Visual fields
- Preauricular lymph nodes
- Systemically well/unwell
- Any facial rash?

Assessment of the red eye

If significant trauma to eye:

Refer immediately

If chemical injury, irrigate continuously with water.

If wears contact lenses:

Refer immediately

Risk of corneal ulceration, microbial keratitis or neovascularization.

Assess acuity with Snellen chart.
Assess each eye separately.
Use a pinhole if glasses at home.

Worry if reduced acuity: almost always warrants urgent referral.
Causes include corneal ulcer/keratitis, anterior uveitis, acute angle glaucoma, scleritis, hyperacute bacterial conjunctivitis.

Is the globe painful?

Worrying! Most painful red eyes need urgent referral.

A corneal abrasion is painful but can be managed in primary care.
Foreign bodies tend to cause irritation but can cause pain: some foreign bodies might possibly be managed in primary care.

Are the pupils reactive to light?

Worry if pupils sluggish/unreactive.

Causes include anterior uveitis and acute angle glaucoma.

Stain cornea to look for corneal abrasions/ulcers.

Corneal ulcers need urgent referral.

Corneal abrasions can be managed in primary care, provided they heal rapidly with appropriate treatment.

Management of the red eye

- Any RED, PAINFUL eye needs same-day referral: see next page.
- Have a very low threshold to refer those who use contact lenses.
- Those with associated systemic disease MAY warrant urgent/same-day referral.
- Some conditions can be managed in primary care: see back page.

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The red eye:
causes which need **SAME-DAY** referral

ALL HAVE A RED, PAINFUL EYE



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Same-day referral usually warranted if red eye and any ONE of the following:

- Acute trauma to the eye.
- Any visual impairment.
- Corneal abnormality (except corneal abrasion: can be managed in primary care).
- Abnormal pupillary reaction.
- Contact lens wear.

Physical abuse can cause a red eye: if suspected, refer through local safeguarding pathways.

ALL HAVE A RED, PAINFUL EYE	Visual acuity affected	Pupil reaction	Photophobia	Circum-corneal injection	Discharge	Abnormal fluorescein stain
Corneal ulcer/keratitis • Contact lens use increases risk. May be viral, bacterial or parasitic.	✓	Normal	✓	No	Yes	✓
Corneal foreign body • If any ferrous metal, refer even if foreign body removed (because rust ring may develop). • Organic material, e.g. seeds/soil/insect associated with increased risk of complications. • For non-ferrous and non-organic foreign body, either use local anaesthetic drops, remove with a cotton bud and follow-up in 24h (if skilled to do so) OR refer, depending on expertise.	Depends on location of FB	Normal	✓	No	Watery	May be visible without staining
Anterior uveitis (iritis) • May see hypopyon (pus in anterior chamber). • Associated diseases: rheumatoid arthritis, spondyloarthropathies, inflammatory bowel disease, reactive arthritis, sarcoidosis, syphilis.	✓	Abnormal	✓	✓	Watery	No
Scleritis • Worse with eye movement, may get headache. • Associated with systemic disease, e.g. RA, SLE, Wegener granulomatosis.	✓	Normal	✓	No	Watery	No
Acute angle glaucoma • Cornea may be hazy, patient sees halos around lights. • Associated headache, nausea/vomiting, abdominal pain. May feel a hard tender globe.	✓	Abnormal	✓	✓	No	No
Hyperacute bacterial conjunctivitis • Sudden onset and rapid progression +/- eyelid swelling. Affects visual acuity, unlike normal bacterial conjunctivitis. Risk of corneal perforation. Can be caused by gonorrhea.	✓	Normal	No	No	++	No
Trauma • Penetrating injury/high-velocity foreign body. • If after hammering/chiselling/working with glass, assume penetrating eye injury until proven otherwise; foreign body may not be visible in primary care.	Maybe	Normal	Maybe	No	Maybe	DO NOT STAIN if penetrating eye injury suspected
Chemical injury • Copious irrigation required to attempt to neutralise the chemical and decrease permanent damage/visual loss.	✓	Normal	✓	No	Maybe	DO NOT STAIN

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The red eye: conditions which can usually be managed in PRIMARY CARE

Red eye: conditions which usually can be managed in PRIMARY CARE

Condition	Symptoms and signs	Management in primary care
Ectropion/entropion	Do not necessarily need referral but: - Entropion can cause a corneal abrasion. - Ectropion can contribute to dry eyes/exposure keratopathy.	For both: - Conservative treatment. - Refer routinely if troublesome symptoms.
Blepharitis	- Irritated eyes, worse on waking. - Itchy and crusted eyelids.	- Warm compress and eyelid cleaning. - Antibiotics, e.g. doxycycline, may be considered if severe/persistent. <i>See separate article in online handbook.</i>
Conjunctivitis	- May be viral, allergic or bacterial (but can be difficult to separate clinically). - Erythema of conjunctiva +/-discharge.	Viral: supportive. Allergic: cold compress, lubricating drops +/- topical antihistamine or mast cell stabiliser (e.g. sodium cromoglicate). Bacterial: supportive +/- topical antibiotics. <i>See separate article in online handbook.</i>
Dry eyes	- Burning, dryness, sensation of foreign body, excess tearing. - Bilateral and chronic. - Think about systemic disease!	Depends on cause: Aqueous insufficiency (reduced secretion from lacrimal glands), can be Sjogren's syndrome-related (autoimmune) or non-Sjogren's syndrome-related (e.g. medication). Evaporative dry eye: dysfunction of lipid layer of tear film resulting in increased tear evaporation. Most often due to meibomian gland dysfunction. <i>See separate article in online handbook.</i>
Corneal abrasion	- Very painful. - Uniform fluorescein uptake with sharp borders. - Caused by trauma or foreign body. (Contact lens user: same-day referral)	- Evert the lids and assess for a retained foreign body. - Treatment: supportive/analgesia, role of antibiotic drops not established. Review after 48h: refer ophthalmology if not better. Eye patch not recommended and may be harmful.
Sub-conjunctival haemorrhage	- Well-demarcated red patch on sclera. - Spares the cornea. - Normal vision and pupillary reaction. - No discharge. - Refer if secondary to trauma.	No treatment required: should resolve in 2–3 weeks. Assess for underlying cause: - Trauma: e.g. blunt eye trauma, foreign body, penetrating injury (refer if penetrating/significant blunt trauma!). - Other: coughing/straining, high BP, bleeding disorder. - Check BP. - If on warfarin, check INR.
Episcleritis	- Limited/isolated patches of injection. - Mild/no pain, mild watering. - Normal pupillary response. - Normal visual acuity.	- Usually self-limiting (lasts up to 3w). - Supportive treatment with artificial tears. - Recurrent episodes: consider underlying systemic disease.

Neonatal conjunctivitis

Consider in first 28d of life:

- REFER! Do not swab in primary care.
- Can cause corneal ulceration (gonorrhoea) or systemic disease/pneumonia (chlamydia).
- Remember: in conjunctivitis, the white of the eye should be red:* if discharge without redness of conjunctiva, is this a blocked lacrimal duct (punctum)?

References: Am Fam Physician 2010;81:137, Am Fam Physician 2016;93:991, BMJ 2010;340:c1711, BMJ 2012;345:e3328, BJGP 2018;68:49, BJGP 2005;55:924, BMJ 2017;359:j4706, DTB 2016;54:9, BJGP 2018;68:49, NEJM 2023;389:2363